Response of a damped system under harmonic excitation of the base

Consider a single degree of freedom damped system whose base is subjected to a hormonic motion

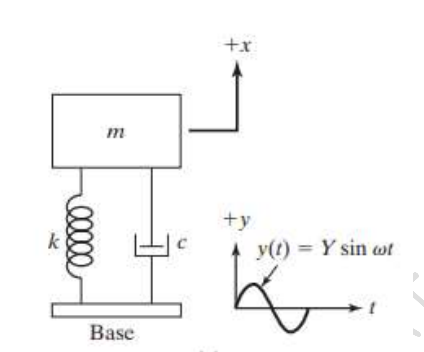
$$y(t) = Y_0 \sin \omega t$$

From the freebody diagram, w can write the equation of motion as

$$m\ddot{x} = -c(\dot{x} - \dot{y}) - k(x - y)$$

$$m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky = c\omega Y_0 \cos \omega t + kY_0 \sin \omega t.$$

Let us substitute $c\omega Y_0 = Y \sin \psi$ and $kY_0 = Y \cos \psi$ so that

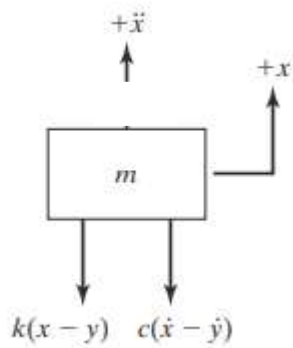


$$Y = Y_0\sqrt{k^2 + (c\omega)^2} = kY_0\sqrt{1 + (2r\zeta)^2} \ll\gg \tan \psi = \frac{c\omega}{k} = 2r\zeta$$

$$m\ddot{x} + c\dot{x} + kx = Y \sin \psi \cos \omega t + Y \cos \psi \ \sin \omega t = Y \sin(\omega t + \psi)$$

$$m\ddot{x} + c\dot{x} + kx = kY_0\sqrt{1 + (2r\zeta)^2} \ \sin(\omega t + \psi)$$

The governing equation is a second order, non homogeneous, linear, ordinary differential equation with constant coefficients. The solution will have two parts; a homogenous solution and a particuar solution.



The homogeneous solution is obtained from the the homogeneous part $m\ddot{x} + c\dot{x} + kx = 0$

Homogeneous solution:

$x_h(t) = (A_1 + A_2 t)e^{-\omega_n t}$, $if\ \zeta = 1$	$x_h(t) = e^{-\zeta\omega_n t}\left(A_1 e^{\omega_n\sqrt{\zeta^2-1}\,t} + A_2 e^{-\omega_n\sqrt{\zeta^2-1}\,t}\right)$, $if\ \zeta > 1$	$x_h(t) = X_0\, e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi)$, $if\ \zeta < 1$

Particular solution:

The particular solution will be of the similar form as $\frac{F_0/k}{\sqrt{(1-r^2)^2+(2r\zeta)^2}} \sin(\omega\, t - \theta)$

$$x_p(t) = \frac{\dfrac{kY_0\sqrt{1 + (2r\zeta)^2}}{k}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega\, t + \psi - \theta) =$$

$$x_p(t) = \frac{Y_0\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega\, t + \psi - \theta) \quad \ll\gg \quad where \quad \tan \theta = \frac{2r\zeta}{1 - r^2}$$

At steady state, the amplitude of response of the system,

$$X = \frac{Y_0\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}}$$

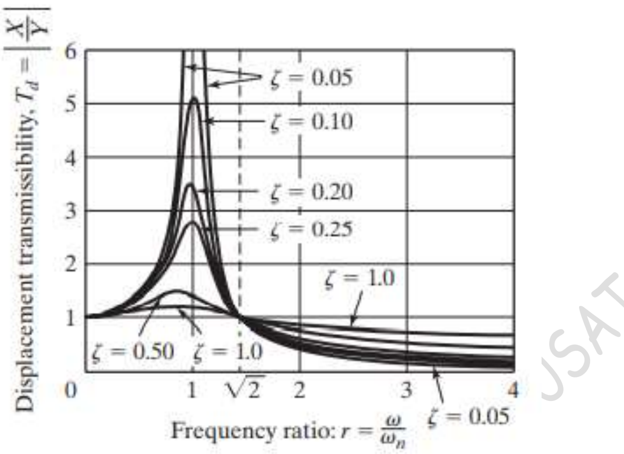
The ratio of the amplitude of steady state response to the amplitude of base motion is known as the displacement transmissibility β_d.

$$\beta_d = \frac{\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}}$$

The figure shows the plot of displacement transmissibility versus frequency ratio.

What are your inferences???

Can you find the value of r when the displacement transmissibility is maximum??



Force transmitted to the base

The force transmitted to the base, $f_T = c(\dot{x} - \dot{y}) + k(x - y) = -m\ddot{x}$

Assuming steady state,

$$x(t) = \frac{Y_0\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega t + \psi - \theta) \quad \lll\ggg \quad \ddot{x}(t) = -\omega^2 \frac{Y_0\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega t + \psi - \theta)$$

$$f_T = -m(-\omega^2)\frac{Y_0\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega t + \psi - \theta = \left(\frac{k}{{\omega_n}^2}\right)\omega^2 \frac{Y_0\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega t + \psi - \theta)$$

$$f_T = \frac{kY_0 \; r^2 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \alpha)$$

The phase angle $\alpha = \theta - \psi \quad \lll\ggg \quad \tan\alpha = \frac{\tan\theta - \tan\psi}{1 + \tan\theta\tan\psi} = \frac{\frac{2r\zeta}{1-r^2} - 2r\zeta}{1 + \frac{2r\zeta}{1-r^2}2r\zeta} = \frac{2\zeta r^3}{1 - r^2 + 4\zeta^2 r^2}$

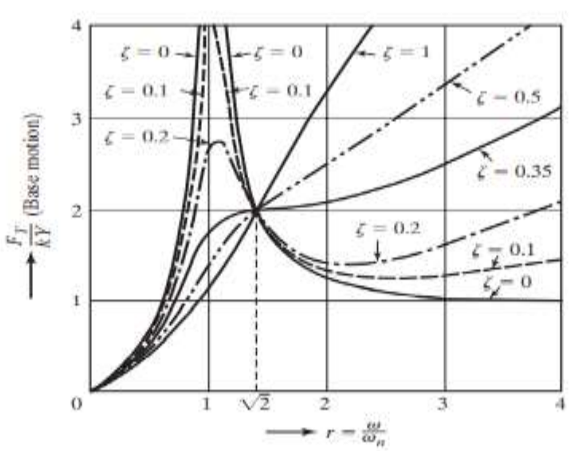
The amplitude of force transmitted to the base at steady state,

$$F_T = \frac{kY_0 \; r^2 \sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}}$$

The ratio of amplitude of dynamic force transmitted to the base to the amplitude of excitation force is known as force transmissibility β_f

$$\beta_f = \frac{F_T}{kY_0} = \frac{r^2\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1 - r^2)^2 + (2r\zeta)^2}}$$

The force transmissibility is plotted against the frequency ratio and is shown .



Relative Motion

Many a cases, the absolute displacement of the mass may not be of interest, but relative displacement of mass with respect to the base may be important. For example in vibration measuring instruments.

Let the relative displacement, $z = x - y$ ⋙ $\dot{z} = \dot{x} - \dot{y}$ ⋙ $\ddot{z} = \ddot{x} - \ddot{y}$ ⋙ $\ddot{x} = \ddot{z} + \ddot{y}$.

The equation $m\ddot{x} = -c(\dot{x} - \dot{y}) - k(x - y)$ can be re-written as

$$m(\ddot{z} + \ddot{y}) = -c\dot{z} - kz \ \ggg\ m\ddot{z} + c\dot{z} + kz = -m\ddot{y}$$

$$m\ddot{z} + c\dot{z} + kz = -m(-\omega^2 Y_0 \sin\omega t) = m\omega^2 Y_0 \sin\omega t$$

The steady state (particular) solution of this equation is of the form $\frac{F_0/k}{\sqrt{(1-r^2)^2+(2r\zeta)^2}}\sin(\omega t - \theta)$

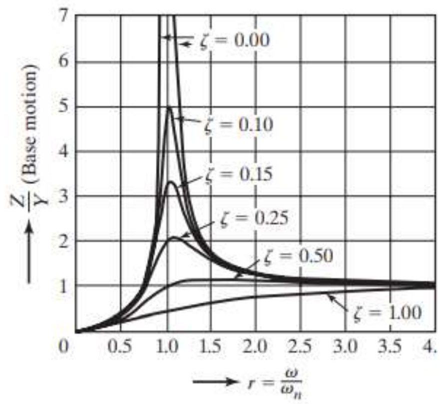
$$z_p(t) = \frac{m\omega^2 Y_0/k}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}\sin(\omega t - \theta) = \frac{r^2 Y_0}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}\sin(\omega t - \theta)$$

$$where \quad \tan\theta = \frac{2r\zeta}{1 - r^2}$$

The amplitude of steady state vibration,

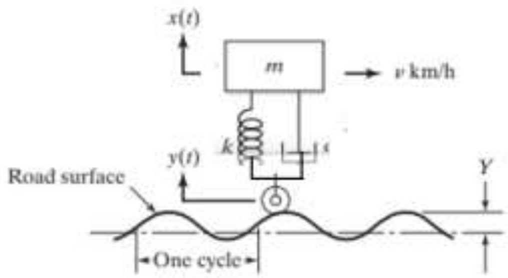
$$Z = \frac{r^2 Y_0}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

$$\frac{Z}{Y_0} = \frac{r^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$



#Example 1

A single defree of freedom model of a motor vehicle that can vibrate in the vertical direction while travelling over a rough road is shown. The vehicle has a mass of $m = 1200\,kg$ and has a suspension system with a spring constant of $k = 400\ kN/m$ and a damping ratio of $\zeta = 0.5$. The speed of the vehicle is $20\ km/hr$, and the road surface is assumed to be a sinusoid with amplitude $Y_0 = 0.05m$ and a wavelength of $6m$. Determine the displacement amplitude of the vehicle.



The speed of the vehicle, $v = 20\frac{km}{hr} = 5.6\ m/s$

Time taken to cover 1m distance$= \frac{1}{5.6}\ s$

Time required to travel one wavelength, $T = \frac{6}{5.6}\ s$

$$\therefore\ \omega = \frac{2\pi}{T} = 5.8\ rad/s$$

$$y = Y_0 \sin\omega t = 0.05\sin 5.8t$$

$$m = 1200\ kg. \gg k = 400000\ N/m$$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{400000}{1200}} = 18.25\ rad/s$$

$$\zeta = 0.5 \ \lll\ggg\ r = \frac{\omega}{\omega_n} = 0.32$$

The steady state amplitude

$$X = \frac{Y_0\sqrt{1 + (2r\zeta)^2}}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = \frac{0.05\sqrt{1 + (2\times 0.32\times 0.5)^2}}{\sqrt{(1 - 0.32^2)^2 + (2\times 0.32\times 0.5)^2}} = 0.055\ m$$

#Example 2

A heavy machine weighing $3000\ N$ is supported on a resilient foundation. The static deflection of the foundation due to the weight of the machine is found to be $7.5\ cm$. It is observed that the machine vibrates with an amplitude of $1\ cm$ when the base of the foundation is subjected to harmonic oscillation at the undamped natural frequency of the system with an amplitude of $0.25\ cm$. Find a) the damping constant of the foundation b) the dynamic force amplitude on the base c)the amplitude of the displacement of the machine relative to the base.

The stiffness of the foundation,

$$k = \frac{W}{\delta} = \frac{3000\ N}{0.075m} = 40000\ N/m$$

$$Frequency\ of\ base\ motion$$
$$= natural\ frequency\ of\ the\ system$$
$$r = 1$$

The natural frequency of the system,

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{40000}{\left(\frac{3000}{9.81}\right)}} = 130.8\ rad/s$$

$$X = 0.01\ m \lll\ggg Y_0 = 0.0025m$$

$$X = \frac{Y_0\sqrt{1+(2r\zeta)^2}}{\sqrt{(1-r^2)^2+(2r\zeta)^2}} \gg\gg 0.01 = \frac{0.0025\sqrt{1+4\zeta^2}}{2\zeta}$$

$$4\zeta^2 = 0.0625(1+4\zeta^2) \gg\gg \zeta = 0.129$$

The damping coefficient of the foundation,

$$c = 2m\zeta\omega_n = 903\ N.s/m$$

The dynamic force amplitude,

$$F_T = \frac{kY_0\ r^2\sqrt{1+(2r\zeta)^2}}{\sqrt{(1-r^2)^2+(2r\zeta)^2}}$$

$$= \frac{4\times10^3\times2.5\times10^{-3}\sqrt{1+4\times0.129^2}}{2\times0.129} = 400\ N$$

Relative displacement amplitude,

$$Z = \frac{r^2Y_0}{\sqrt{(1-r^2)^2+(2r\zeta)^2}} = 0.00968\ m$$

Exercise 1

The base of a damped spring-mass system with $m = 25\ kg$ and $k = 2500\ N/m$, is subjected to a harmonic excitation $y(t) = Y_0 \sin\omega t$. The amplitude of the mass is found to be $0.05\ m$ when the base excited at the natural frequency of the system with $Y_0 = 0.01\ m$. Determine the the damping constant of the system.

Exercise 2

Determine the steady state response of the mass of a spring-mass-damper system subjected to a harmonic base excitation $y(t) = 0.001\sin 400t$. Take $m = 1\ kg, k = 50000\frac{N}{m},\ c = 50\ Ns/m$

Vibration Measuring Instruments

The expression for relative displacement of the mass with respect to the base is given by

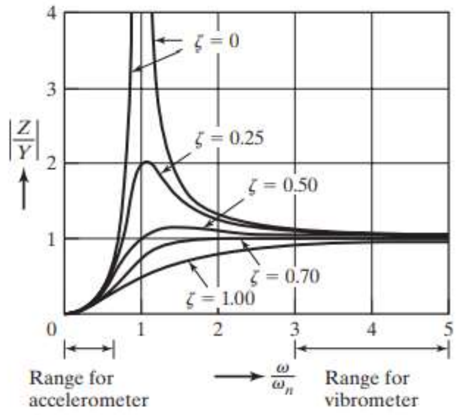
$$z(t) = \frac{r^2 Y_0}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \sin(\omega t - \theta)$$

Vibrometer

Vibrometer or seizmometer measures the displacement of the vibrating body. The amplitude of relative displacement at steady state,

$$Z = \frac{r^2 Y_0}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

$$Z = Y_0 \quad if \ \frac{r^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = 1$$



#Example 1

A vibrometer having a natural frequency of $4\,rad/s$ and $\zeta = 0.2$ is attached to a structure that performs a harmonic motion. If the difference between the maximum and the minimum recorded values is $8\,mm$, find the amplitude of motion of the vibrating structure when its frequency is $40\,rad/s$.

Amplitude of $Z = 4\,mm,\ \zeta = 0.2,\ \omega_n = 4\frac{rad}{s},\ \omega = 40\frac{rad}{s},\quad r = 10$

$$\frac{r^2}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = 1.0093$$

$$\therefore\ Y_0 = \frac{Z}{1.0093} = 3.9631\,mm$$

Accelerometer

The base motion is reperesented by

$y(t) = Y_0 \sin\omega t$

The acceleration of the base, $\ddot{y} = -\omega^2 Y_0 \sin\omega t$

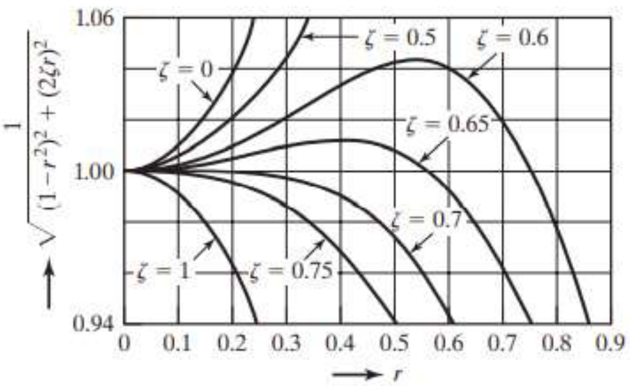
The amplitude of acceleration= $-\omega^2 Y_0$

Now we have,

$$Z = \frac{r^2 Y_0}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = \frac{\left(\frac{\omega}{\omega_n}\right)^2 Y_0}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

$$-\omega_n^2 Z = -\omega^2 Y_0$$

if $\frac{1}{\sqrt{(1-r^2)^2+(2r\zeta)^2}} = 1$



#Example 1

An accelerometer has a suspended mass of $0.01\,kg$ with a damped natural frequency of vibration of $150\,Hz$. When mounted on an engine undergoing an acceleration of $1g$ at an operating speed of $6000\,rpm$, the acceleration recorded as $9.5\,m/s^2$ by the instrument. Find the damping constant and the spring consant of the accelerometer.

Measured value of acceleration($\omega_n^2 Z$) $= 9.5\,m/s^2$ Actual value of acceleration($\omega^2 Y_0$) $= 9.8\,m/s^2$

$$0.01\,kg \quad \lll\ggg \quad \omega = \frac{2\pi N}{60} = 628.32\frac{rad}{s}, \qquad \lll\ggg \qquad \omega_d = 2\pi f = 942.48\,rad/s$$

$$\omega_n^2 Z = \omega^2 Y_0 \times \frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}}$$

$$9.5 = 9.8 \times \frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} \quad \ggg \quad \frac{1}{\sqrt{(1-r^2)^2 + (2r\zeta)^2}} = 0.9684$$

$$(1-r^2)^2 + (2r\zeta)^2 = 1.0642$$

$$\frac{\omega}{\omega_d} = \frac{628.32}{942.48} = 0.667 = \frac{\omega}{\omega_n\sqrt{1-\zeta^2}} = \frac{r}{\sqrt{1-\zeta^2}}$$

$$r^2 = 0.444(1-\zeta^2)$$

$$\left(1 - 0.444\left(1-\zeta^2\right)\right)^2 + 0.444\left(1-\zeta^2\right)4\zeta^2 = 1.0642$$

$$1.580\zeta^4 - 2.274\zeta^2 + 0.7576 = 0 \quad \ggg \quad \zeta = 0.7253, 0.9547$$

Choosing $\zeta = 0.7253$

$$\omega_d = \omega_n\sqrt{1-\zeta^2} \ggg\gg 942.48 = \omega_n\sqrt{1-0.7253^2} \ggg \omega_n = 1369.02\,rad/s$$

$$k = m\omega_n^2 = 0.01 \times 1369.02^2 = 18742.2\,N/m$$

$$c = 2m\zeta\omega_n = 2 \times 0.01 \times 0.7253 \times 1369.02 = 19.86\,Ns/m$$

Exercise 1

A spring-mass system with $m = 0.5\,kg$ and $k = 10000\,N/m$, with negligible damping, is used as a vibration pickup. When mounted on a structure vibrating with an amplitude of $4mm$, the total displacement of the mass of the pickup is observed to be $12\,mm$. Find the frequency of the vibrating structure.

Exercise 2

A spring-mass-damper system, having an undamped natural frequency of 100 Hz and a damping constant of 20 Ns/m, is used as an accelerometer to measure the vibration of a machine operating at a speed of 3000 rpm. If the actual acceleration is 10 m/s^2 and the recorded acceleration is 9 m/s^2, find the mass and spring constant of the accelerometer.

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition